Metamorphosis

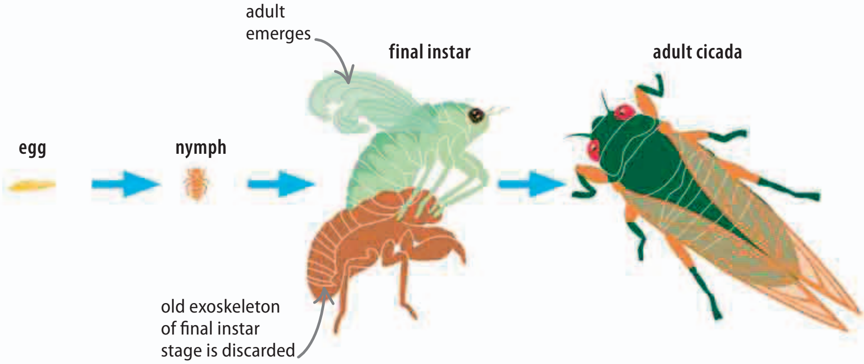
Animals that produce large numbers of young can find they are in direct competition for food with their own offspring. Many insects avoid this problem by having larval stages, which look and live in very different ways to the adults. A larva must undergo a complete metamorphosis, where its body rebuilds itself in the adult, sexually mature form. Other young insects are nymphs, which, unlike larvae, resemble their adult form.

REAL WORLD

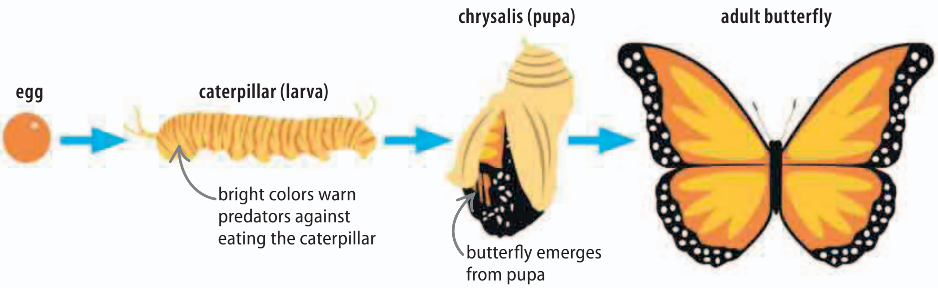
Woolly bear caterpillar

The larvae of tiger moths are called woolly bears, and the species that live in the Arctic take years to reach adulthood. The woolly bears freeze solid during the long winters, and can only manage one molt during the short Arctic summer. After 14 molts and 14 years, the caterpillars finally pupate into tiger moths.

▷ **Incomplete metamorphosis**
The cicada nymph looks like its adult parent, but lacks wings. After several molts, the nymph reaches its largest size, called the final instar. During the next molt, it develops wings and sex organs and emerges as an adult, ready to reproduce.



▷ **Complete metamorphosis**
After hatching, the caterpillar (larva) is an eating machine and undergoes several molts as it outgrows its inflexible exoskeleton. Then it becomes a pupa, a dormant phase inside a protective case, where metamorphosis takes place. The insect emerges as an adult butterfly.



Reproductive strategies

Animals employ different strategies to ensure their offspring survive until they can reproduce. There are two main options: producing huge numbers of young, but leaving their survival to chance, or protecting just a few young, and giving them parental care and protection.

▷ **Pros and cons**
All reproductive strategies have advantages and disadvantages. An animal's place in the food chain and its habitat are the two factors that influence its reproductive strategy.

Animal	Type of care	Benefits	Costs
salmon	many thousands of eggs are laid each year	young can populate a new habitat quickly, and at least a few will always survive	effort kills the parents, and most young die before they reproduce
lion	one or two young are produced every few years; mother looks after them until adulthood	young are more likely to survive until adulthood, and help raise and protect younger siblings	investing energy into just a few young over many years is risky